LIBBY CASE REPORT

demo-nsclc-egfr-l858r-post-osi-d3m0

Multi-agent decision support — clinician track

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Executive summary

Case:demo-nsclc-egfr-l858r-post-osi-d3m0**Question:**EGFR L858R-mutant stage IV lung adenocarcinoma with on-target MET amplification (GCN 8.2) after 22 months of first-line osimertinib. T790M is absent. What's the next line, given a working artist who needs her hands and prefers trials and oral therapy? **Evidence base:**4 trials, 5 clinical-evidence rows, 4 preclinical rows. Three ranked interventions. Board agreement scores span +0.8 to -0.4, so rank 1 carries near-consensus and rank 2 carries an active veto.

What this report covers

The PI's synthesis of the post-osimertinib decision when MET amplification is the dominant resistance driver. There are no scenario branches in this case; the three ranked options are unconditional, and the gap between rank 1 and rank 2 is large enough that the report reads as one main recommendation plus two transparency entries.

Top-line findings

- MET amplification at GCN 8.2 by FISH, with T790M absent, points cleanly at a MET-directed combination. Savolitinib plus osimertinib is the option that fits both the biology and the patient's stated preferences.
- The board endorsed rank 1 four-to-one. The concensusite flagged that savo-osi is not yet inside NCCN; the resolution is to enroll on SAFFRON or SACHI rather than treat off-trial, which the patient already prefers.
- The conservative vetoed rank 2 (amivantamab + lazertinib) on severe-neuropathy and IV-burden grounds. Critic and advocate also dissented. Two personas endorsed it on guideline alignment and effect size, so it stays on the table as considered_with_caveatsrather than being dropped — but override of the veto is not recommended unless rank 1 fails.
- Rank 3 (patritumab deruxtecan) triggers the patient's alopecia veto outright. Only the risktaker advanced it. It's listed for transparency, not as a live option.

Recommendation summary

1. **Savolitinib + osimertinib (preferably on SAFFRON / SACHI)**— recommended. ORR ~30% in the MET-amp post-EGFR-TKI stratum (TATTON expansion, PMID 32679432). All-oral; clears every stated veto; trial-route resolves the guideline gap.
2. **Amivantamab + lazertinib**— considered with caveats. ORR 36% in the CHRYSALIS-2 post-osimertinib subset. Conservative veto holds on neuropathy and IV chair time, both of which matter unusually for a patient whose work depends on her hands.
3. **Patritumab deruxtecan (HER3-DXd)**— considered with caveats. Triggers the alopecia veto. Surfaced so the user can revisit it with her oncologist if she chooses to re-weight that veto.

What this report does *not* cover

Dose adjustments, monitoring schedules, sequencing across later lines, payer or access logistics, or institution-specific trial slot availability. Those belong to the treating team.

How to use this report

Each ranked option carries the board's agreement state (endorsed, dissent, veto by persona) and one to three anchor citations. The full per-trial extraction lives on the live case page; this PDF is the synthesis.

Libby is an experimental decision-support tool. The recommendations on this page have not been reviewed by a clinician treating this patient. Do not act on this report without consulting a qualified oncologist.

<!-- libby:downloads:begin -->

Downloads

- Clinician PDF report— ranked recommendations + evidence + sources
- Patient/caregiver PDF— plain-language summary
- Master manuscripts table (PDF)— every paper considered — n, effect, variance, toxicities
- Self-contained HTML— recommendations table that opens offline

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Research question

In stage IV EGFR L858R-mutant lung adenocarcinoma after 22 months of first-line osimertinib (PR), with on-target MET amplification (GCN 8.2 by FISH) and T790M absent, what next-line interventions could target the resistance mechanism — given a working artist who needs her hands and prefers oral therapy plus trials?

Patient profile (scrubbed)

- **Primary site / histology:**lung adenocarcinoma
- **Stage:**IV
- **Performance status:**ECOG 1
- **Age band:**60-69
- **Biomarkers:**EGFR L858R (present); EGFR T790M (absent); MET amplification (GCN 8.2 by FISH); PD-L1 TPS 10%
- **Prior therapy:**1L osimertinib, PR x 22 months — now progressed
- **Targetable features:**EGFR L858R (sensitizing); MET amplification (likely on-target resistance)

Preferences

- **Efficacy/toxicity weight:**0.55 (slight efficacy lean)
- **Toxicity vetoes:**severe neuropathy, cardiotoxicity, alopecia
- **Modality constraints:**oral preferred; no inpatient infusion; minimize IV chair time
- **Free text:***"working artist — manual dexterity in hands matters more than typical"*
- **Trials preferred:**yes

Scope summary

4 trials, 5 clinical-evidence rows, 4 preclinical rows. Three ranked interventions; no scenario branching (every biomarker is confirmed). Board agreement scores span +0.8 (rank 1) to -0.4 (rank 2, with an active veto), so the rank-1/rank-2 gap is large and load-bearing.

Cross-cutting caveat (read first)

MET amplification at GCN 8.2 with T790M absent points cleanly at MET-directed combination therapy, and the patient's modality + toxicity preferences foreclose the most-cited alternative. This shapes every rank below.

- T790M-negative status removes second-generation osimertinib re-challenge as a sequencing option; the resistance is not gatekeeper-mutation-driven.
- MET amplification is the on-target resistance mechanism in roughly 15–30% of post-osimertinib progressors at this GCN range; savolitinib + osimertinib is the mechanism-matched combination with replicated evidence in this exact subset.
- The user's manual-dexterity-driven veto on **severe neuropathy**combined with **minimize IV chair time**removes amivantamab + lazertinib from contention as rank 1 even though it carries NCCN cat-2A status. The conservative's toxicity veto on this rec is the load-bearing dissent.
- The **alopecia veto**alone forecloses patritumab deruxtecan independent of efficacy. The dossier preserves it for transparency, not as a live option.

Intervention grouping

- **MET-targeted EGFR-TKI combinations (mechanism-matched, all-oral):**savolitinib + osimertinib via the SAFFRON / SACHI phase-3 program. Mechanistic anchor: TATTON expansion (PMID 32679432).
- **EGFR + MET bispecific + 3G TKI (broader-spectrum, IV-anchored):**amivantamab + lazertinib via CHRYSALIS-2 (PMID 36720074).
- **HER3-directed antibody-drug conjugate (mechanism-agnostic post-TKI option):**patritumab deruxtecan via HERTHENA-Lung01 (PMID 37563559).

Top interventions

Rank 1. Savolitinib + osimertinib (preferably on SAFFRON / SACHI)

Mechanism-matched, all-oral, clears every preference axis. Four of five personas endorse; concensusite's qualified guideline-fit concern resolves on trial enrollment.

Evidence base

The mechanistic and clinical anchor is the TATTON expansion (PMID 32679432; n=69; ORR 30%, 95% CI 19–42 in the EGFR-mut + MET-amp post-EGFR-TKI subset). The phase-3 confirmatory program (NCT05261399; SAFFRON; n=324; savo + osi vs platinum + pemetrexed; PFS endpoint) is recruiting and reads out into the same biomarker-defined population. The patient's MET-amp GCN 8.2 by FISH exceeds the GCN ≥6 / IHC 3+ enrollment threshold cleanly.

Likelihood of desired effect

High mechanism concordance: MET amplification IS the on-target resistance mechanism this combination targets, and the patient's biomarker fits the studied stratum exactly. The TATTON ORR (~30%) is real, replicated within program, and pre-specified to this subset. The phase-3 readout will quantify durability and PFS gain over chemotherapy.

Toxicity profile

- LFT elevation (manageable with dose reduction)
- Peripheral edema
- Fatigue
- All-oral; no inpatient component
- No severe-neuropathy signal in either TATTON or SAFFRON program reports

User vetoes (severe neuropathy, cardiotoxicity, alopecia): all clear. Modality constraints (oral, no inpatient, minimize IV): all clear.

Counter-productive mechanisms / dissent

Endorsed by risktaker, conservative, critic, and advocate. Concensusite raised a *qualified guideline-fit*concern: this combination is not currently NCCN-listed as a recommended regimen, so off-trial use is off-label. The resolution is trial enrollment, which is itself preferred by the patient. No persona dissented or vetoed.

Practical considerations

- All-oral: savolitinib 300 mg PO BID + osimertinib 80 mg PO daily
- Trial enrollment route is the guideline-aligned path
- Off-trial use is off-label — surface this with the treating team
- LFT monitoring at baseline + every 2–4 weeks early in therapy

Why this rank

Strong combined evidence-and-preference fit. Every preference axis clears; mechanism matches the biomarker; agreement_score (+0.8) is the highest in the dossier. The rank-1/rank-2 gap is +1.2 (vs amivantamab + lazertinib at -0.4), which reflects both the active veto on rank 2 and the better preference fit here.

Per-trial detail

Therapeutic agent	Efficacy	Toxicity	Reference
Savolitinib + osimertinib — TATTON expansion (single-arm phase 1b, n=69 EGFR-mut + MET-amp post-TKI)	ORR 30% (95% CI 19–42)	LFT elevation, peripheral edema, fatigue (all manageable)	PMID 32679432
Savolitinib + osimertinib — SAFFRON (phase 3 RCT, recruiting, n=324)	PFS endpoint; outcome pending	Per TATTON safety profile + comparator chemo arm	NCT05261399

Rank 2. Amivantamab + lazertinib

*Real efficacy signal, NCCN cat-2A, but **conservative vetoed on neuropathy + IV-burden grounds**. Considered with caveats; override not recommended absent rank-1 failure.*

Evidence base

CHRYSALIS-2 (PMID 36720074; single-arm phase 1/2; n=162; post-osimertinib EGFR-mutant biomarker subset including MET-amp): ORR 36% (95% CI 28–45). NCCN NSCLC v3.2025 category 2A; ESMO MCBS B.

Likelihood of desired effect

Effect size is real and slightly larger than rank 1's ORR. Biomarker fit is acceptable (EGFR L858R + MET-amp permitted), though the trial isn't enriched specifically for MET-amplified post-osimertinib progressors at this GCN range.

Toxicity profile

- **Severe neuropathy**in combination data — triggers user veto
- IRR grade 3+ ~7%
- Rash
- Inpatient or extended chair-time IV cycle 1 — triggers modality preference conflicts (no inpatient infusion; minimize IV chair time)

For a working artist whose manual dexterity matters more than typical, the neuropathy risk is decision-relevant beyond the categorical veto.

Counter-productive mechanisms / dissent

Conservative issued a toxicity veto on stated severe-neuropathy and IV-burden grounds. Advocate dissented on preference fit. Critic dissented on evidence-quality grounds (single-arm phase 1/2, no randomized comparator in the post-osimertinib subset). Risktaker endorsed on effect size; concensusite endorsed on NCCN guideline alignment.

The veto is honored, not silently dropped: per PI synthesis rules, this row stays on the page as considered_with_caveatsso the user sees what was considered. Override of the conservative's veto is **not** recommended absent failure of the rank-1 option.

Practical considerations

- amivantamab 1050 mg IV weekly cycle 1 then q2w + lazertinib 240 mg PO daily
- Subcutaneous amivantamab formulation in development; AE re-profiling pending
- NCCN-recommended; off-label considerations are weaker than for rank 1

Why this rank

Below savolitinib + osimertinib because of the active toxicity veto plus the modality conflict, despite the slightly larger ORR. Above patritumab deruxtecan because two personas affirmatively endorsed and the alopecia veto on rank 3 is unconditional.

Per-trial detail

Therapeutic agent	Efficacy	Toxicity	Reference
Amivantamab + lazertinib — CHRYSALIS-2 (single-arm phase 1/2, n=162 post-osimertinib)	ORR 36% (95% CI 28–45)	Severe neuropathy (combination); IRR G3+ ~7%; rash	PMID 36720074

Rank 3. Patritumab deruxtecan (HER3-DXd)

*Real efficacy, but **the alopecia veto forecloses it.**Surfaced only so the user can re-weight the veto with her oncologist if she chooses.*

Evidence base

HERTHENA-Lung01 (PMID 37563559; single-arm phase 2; n=225; EGFR-mutant NSCLC after EGFR-TKI): ORR 29.8% (95% CI 23.9–36.2). Effect size is comparable to ranks 1 and 2.

Likelihood of desired effect

The mechanism (HER3-targeted antibody-drug conjugate) is mechanism-agnostic relative to the patient's MET-amp resistance — it targets HER3 expression rather than the resistance pathway. Effect is real but mechanism-fit is weaker than rank 1.

Toxicity profile

- ILD grade 3+ ~5%
- **Alopecia**— triggers user veto
- Thrombocytopenia
- IV q3w

Counter-productive mechanisms / dissent

Only the risktaker advanced this rec. Advocate did not advance it because the alopecia veto was stated. No formal veto was issued because the dossier surfaced the rec only after ranks 1 and 2 were established.

Practical considerations

- 5.6 mg/kg IV q3w
- Not currently NCCN-endorsed as standard for this setting
- ILD risk requires baseline pulmonary function + monitoring

Why this rank

Below ranks 1 and 2 because the alopecia veto is unconditional. Surfaced for transparency so the user can ask her oncologist whether re-weighting the alopecia veto changes the calculus — that's a conversation, not a recommendation.

Per-trial detail

Therapeutic agent	Efficacy	Toxicity	Reference
Patritumab deruxtecan — HERTHENA-Lung01 (single-arm phase 2, n=225 post-EGFR-TKI)	ORR 29.8% (95% CI 23.9–36.2)	ILD G3+ ~5%; alopecia; thrombocytopenia	PMID 37563559

Classes examined but not ranked

- **Osimertinib re-challenge / dose escalation:**T790M absent — gatekeeper-mutation pathway is not the resistance mechanism, so re-challenge has no biomarker rationale.
- **MET-targeted monotherapy (capmatinib, tepotinib):**mechanistically targets the same feature class as the rank-1 savolitinib + osimertinib combo, but the approved indications are METex14 not amplification — biomarker subset mismatch.

Ranked prioritization

Rank	Status	Intervention	Endorsed by	Dissent	Veto	Likelihood	Toxicity burden	Why this rank
1		Savolitinib + osimertinib (preferably on SAFFRON / SACHI)	<span <span <span <span	—	—	High mechanism fit; ORR ~30% replicated	LFT, edema, fatigue	Mechanism + preferences both clean
2		Amivantamab + lazertinib	<span <span	<span <span	<span	ORR 36% in subset	Severe neuropathy, IV burden	Active veto + modality conflict
3		Patritumab deruxtecan (HER3-DXd)	<span	—	—	ORR 29.8% post-TKI	ILD ~5%; alopecia	Triggers alopecia veto outright

Caveats

- **Evidence base is modest in size.**Rank 1's anchor is a 69-patient TATTON expansion plus a recruiting phase 3; rank 2's anchor is a 162-patient single-arm; rank 3's anchor is a 225-patient single-arm. None has a randomized comparator in the patient's exact biomarker stratum yet.
- **No OS data**for any of the three ranked interventions in the post-osimertinib MET-amp setting specifically. Effect estimates are ORR-based; durability is the open question for all three.
- **Patient-context veto sensitivity.**Two of three rec-ranks turn on user vetoes (neuropathy, alopecia) tied to the patient's working-artist context. The board treated these as load-bearing, not disposable.
- **What would change the ranking:**
    - SAFFRON OS readout, if it confirms a meaningful PFS / OS benefit, would tighten rank 1's confidence and may push it from "preferred" to "guideline-fit."
    - A MET-amp-stratified amivantamab + lazertinib randomized readout that mitigates neuropathy (e.g. with the SC formulation) could narrow the rank-1 / rank-2 gap, but the modality preference would still anchor rank 1 above it.
    - User re-weighting of the alopecia veto would move patritumab deruxtecan from rank 3 (considered_with_caveats) to rank 2 if the user accepted the alopecia tradeoff.
    - Failure of the rank-1 option (intolerance, progression, slot unavailability) is the explicit precondition for revisiting rank 2 with the conservative's veto on the table.
- **Re-scoping caveat.**If the patient's clinical state changes (e.g. CNS progression, declining ECOG), or if the modality preference relaxes, the ranking shifts. The current page assumes the stated profile and preferences

hold.

Sources

PubMed (PMID):

- 32679432— Sequist et al., TATTON expansion, *Lancet Oncol*2020
- 36720074— Cho et al., CHRYSALIS-2, *JCO*2023
- 37563559— Yu et al., HERTHENA-Lung01, *JCO*2023

ClinicalTrials.gov (NCT):

- NCT05261399— SAFFRON (savolitinib + osimertinib phase 3)
- NCT04077463— CHRYSALIS-2 (amivantamab + lazertinib)
- NCT04619004— HERTHENA-Lung01 (patritumab deruxtecan)
- NCT02143466— TATTON

Transparency artifacts

- Trial table— 4 trials in the dossier
- Evidence list— 5 clinical-evidence rows + 4 preclinical rows
- Tumor-board transcript— 5 positions, 20 cross-critiques
- Recommendations table— full ranked detail
- Plain-language summary— patient/caregiver track

Run log

Demo case authored May 2026 with synthetic data for smoke-testing the pipeline. No real patient information is represented. Re-rendered to shieldbreak-flavored layout in the same month after Libby's PI authoring contract was updated to mirror pirl-unc/io-shieldbreak.

Sources

PubMed (PMID):

- 32679432
- 36720074
- 37563559

ClinicalTrials.gov (NCT):

- NCT05261399